# GraniteMeasure AI

10-minute demo script - AI-based granite block measurement and seigniorage assessment

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**How to read this page:** SAY *Yellow box = say this aloud.* DO **Grey box = tap or do this.**

| Time | Part | What you show |
|---|---|---|
| **0:00** | Introduction | Who you are and what the app does |
| **0:30** | Home page | Dashboard, today's blocks, flags |
| **1:30** | Blocks page | Ledger, CSV and OMEPS export |
| **2:15** | Live measurement | Five photos, marker, Measure block |
| **5:15** | Results | Useful block, 3D view, faces, tape check |
| **8:15** | Method | Marker, homography, AI, geometry |
| **9:15** | Parameters and critical cases | Open source stack, limits |
| **9:45** | Close | Verify page and OMEPS export |

## 1. Introduction 0:00 - 0:30

SAY *Good morning. I am Thiru from Haryak Technologies. We built GraniteMeasure AI: a mobile app that measures a granite block from photos, works out the useful block after cracks and dressing, classifies the seigniorage category and slab, and checks the result against weighbridge and dispatch records.*

## 2. Home page 0:30 - 1:30

DO **Open the Home screen.**

SAY *This is the home page for a quarry operator. It shows today's blocks, total volume, the estimated seigniorage, and any weighbridge mismatches that need inspection. Every block carries GPS, a timestamp, the quarry and an ID.*

## 3. Blocks page 1:30 - 2:15

DO **Tap Blocks.**

SAY *This is the ledger. Each block can be searched and filtered by category, exported as CSV for reporting, and exported as a record for OMEPS. Every record carries a digital fingerprint, so it cannot be changed without detection.*

## 4. Live measurement 2:15 - 5:15

DO **Tap Measure. Enter the block ID. GPS and time are captured automatically.**

SAY *Now a live measurement. Five photos, one per face: front, back, left, right and top. The bottom cannot be photographed, so it is taken to match the top.*

DO **Take the 5 photos with the live camera. Lay the printed marker flat on each face.**

SAY *For accuracy, the operator lays this printed marker, a known 40 centimetre size, flat against each face. The camera checks it live and tells us if it is too small or at a bad angle.*

DO **Tap Measure block. Wait 20 to 40 seconds.**

SAY *While it analyses: this is the same process an officer follows with a tape, but standardised and recorded.*

## 5. Results, one by one 5:15 - 8:15

### 1) Detected state and confidence

SAY *It first decides whether the block is raw or dressed, and states its confidence and error range honestly.*

### 2) Total useful block

SAY *Final result: read the length, width and height on screen, then the volume in cubic metres and the weight in tonnes. Name the category and the slab, then the indicative seigniorage using the rates in force.*

### 3) Visualisation

SAY *Here is the raw block in 3D. The red zones are cracks or weak areas found on the photos. The green block is the largest clean block that avoids them. Below are the front, side and top views with dimensions.*

### 4) Face-by-face inspection

SAY *Each photo shows the defects found, how much of that face is useful, and why.*

### 5) Tape check (if you have one)

SAY *Here the officer types the real tape measurement, and the app shows the exact difference for each dimension.*

**Read the numbers from your own screen.** Say the length, width and height of the useful block, then the volume, weight, category, slab and indicative seigniorage exactly as shown. Length is the longest side, width the other horizontal side, height vertical. Only present results from a real measurement with a working API key.

## 6. How the measurement works 8:15 - 9:15

**SAY** *Three steps.*

**SAY** ***One, scale.*** *The printed marker is an ArUco marker, detected with the open-source js-aruco2 library, which uses OpenCV's marker standard. Its four corners give a homography, which turns pixels on that face into real centimetres and corrects the camera angle.*

**SAY** ***Two, the face outline and the defects.*** *A vision AI model, Claude, traces each face's outline and finds cracks, holes and weak zones, because that part needs recognition. Our code then converts the outline into centimetres using the marker.*

**SAY** ***Three, combining.*** *Length, width and height are the median over the faces that show them. A crack seen on one face is carried through the whole block, and the app finds the largest clean rectangular block that avoids every bad zone. For a raw block it also removes the dressing allowance.*

## 7. Parameters, open source and critical cases 9:15 - 9:45

**SAY** *The parameters are defect type, severity, extent with a safety margin, the dressing allowance, density for weight, and the gangsaw slab thresholds of 270 by 150 centimetres.*

**SAY** *The production version uses an open-source stack: OpenCV, COLMAP for 3D reconstruction, Open3D, SAM 2 for segmentation and YOLO for defect detection.*

**SAY** *The critical cases are irregular raw blocks, hidden internal cracks that no photo can show, shadows and dust, and photos taken at a steep angle. Without a marker the app falls back to an estimate and labels it with a wider error range. We do not hide that.*

## 8. Close 9:45 - 10:00

**SAY** *Finally, the Verify page cross-checks the AI weight against weighbridge and dispatch, flags mismatches beyond tolerance, and exports an OMEPS-ready record.*

## Open-source stack reference

| Tool | Licence | Use |
|---|---|---|
| OpenCV + ArUco markers | Apache-2.0 | Exact scale and perspective correction |
| js-aruco2 (in the app today) | MIT | Marker detection in the browser |
| COLMAP, OpenMVG/OpenMVS, Meshroom | BSD / MPL-2.0 | 3D reconstruction from the photos |
| Open3D | MIT | Bounding box, dimensions, volume |
| Depth Anything V2, MiDaS | Apache-2.0 / MIT | Depth from one photo (relative only) |
| SAM 2 | Apache-2.0 | Block and face segmentation |
| YOLO, Detectron2 | AGPL-3.0 / Apache-2.0 | Crack, hole and void detection |
| PyTorch + timm | BSD / Apache-2.0 | Colour category classification |

Check each licence (for example AGPL and non-commercial ones) before government deployment.

## Questions to be ready for

### Q: Is this 100% accurate?

SAY *No. Accuracy depends on the marker and the photo. We show the error range on every result, and an officer can still check with a tape.*

### Q: Is it open source?

SAY *The scale and geometry are. The face tracing uses a commercial AI model today, and the production plan replaces it with open-source models.*

### Q: Why does the same photo give the same answer?

SAY *Results are rounded and stored, so the same photos always give the same result.*

### Q: Seigniorage on raw or dressed volume?

SAY *The app currently uses the dressed block. It is a one-line change if the department rules otherwise.*

## Before you start

• Print the marker beforehand: Print the scale marker, on the Home screen. Check the black square with a tape.

• Have a tape ready for the tape check.

• Bring one block with its five photos already taken, in case the quarry network is slow.

• Use the HTTPS link on the phone so live camera and GPS work.

• Do not quote any accuracy percentage you have not measured on real blocks.